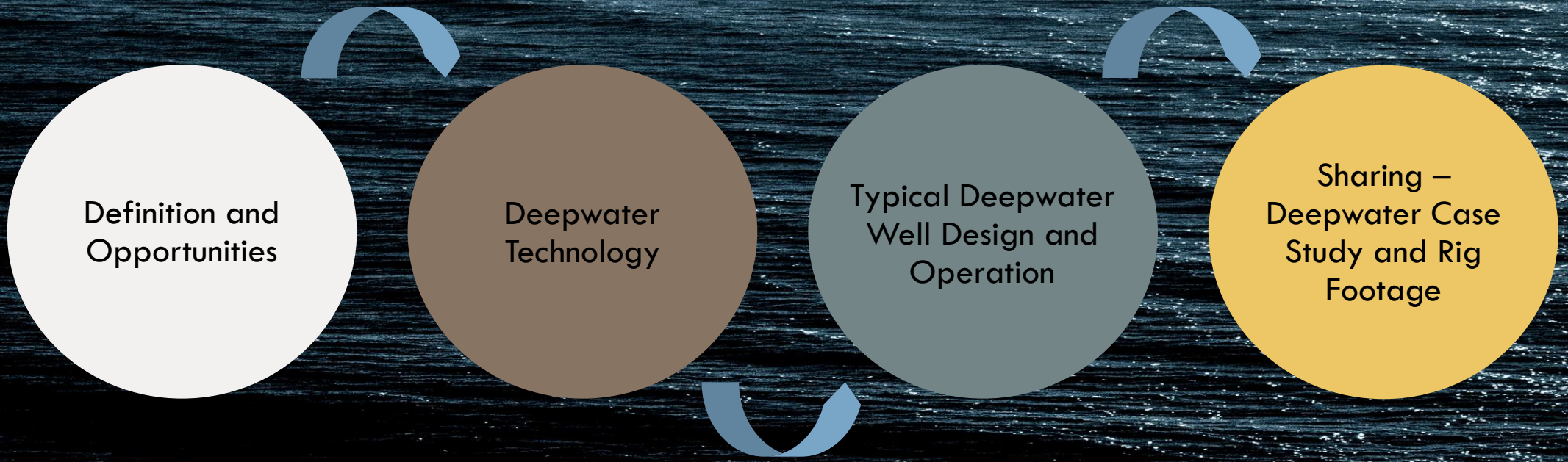
Introduction to Deepwater Operation

JDS

30 October 2026

Outline

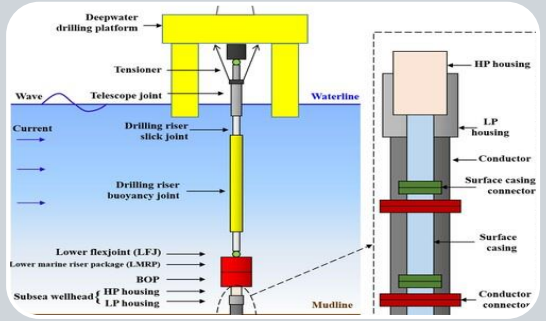

Definition and Opportunities
Deepwater Technology
Typical Deepwater Well Design and Operation
Sharing – Deepwater Case Study and Rig Footage

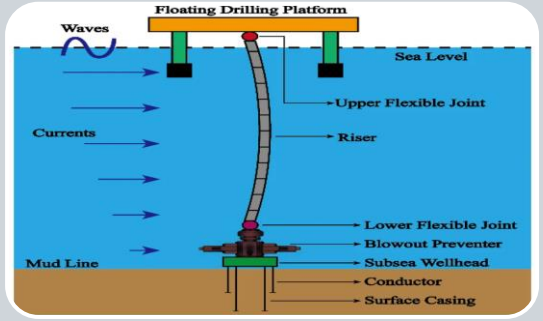
TRANSITION: ONSHORE TO DEEPWATER

Evolution of Oil & Gas Operations: From Surface to Subsea Complexity









Onshore

Accessible, lower cost

Direct well intervention

Limited logistical complexity

Offshore (Shallow Water)

Platform-based operations

Weather & marine exposure

Increased logistics dependency

Subsea / Deepwater

No surface facilities

Fully remote operation (ROV, subsea systems)

High cost, high risk, high integration

As we move deeper, **complexity increases exponentially—not linearly.**

Deepwater Definition

The process of drilling holes in the ocean seabed of the continental shelf and also applies to drilling in lakes and inland seas. The first offshore drilling activities date back to the late 19th century in inland waters of Grand Lake St. Marys, Ohio.

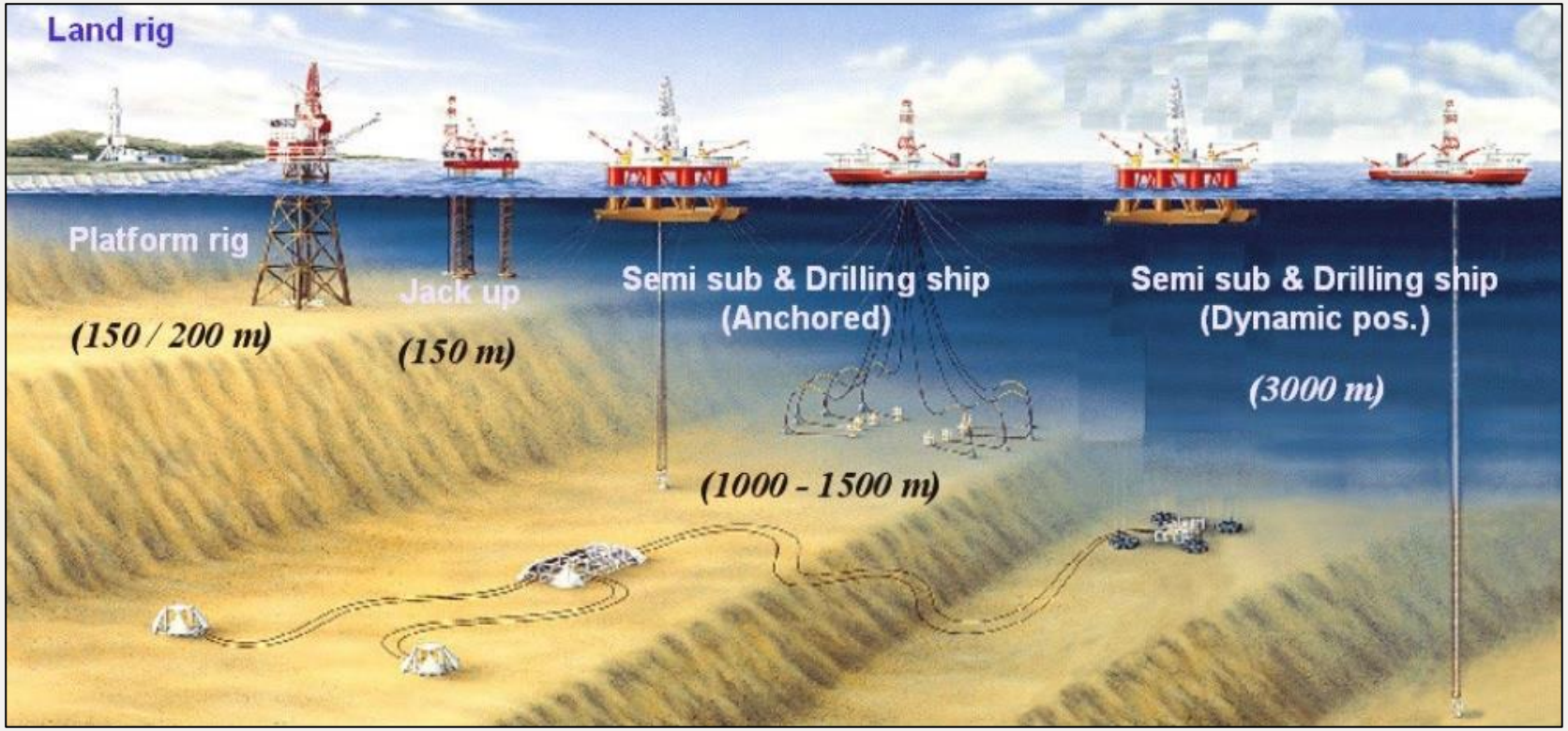
Shallow Water

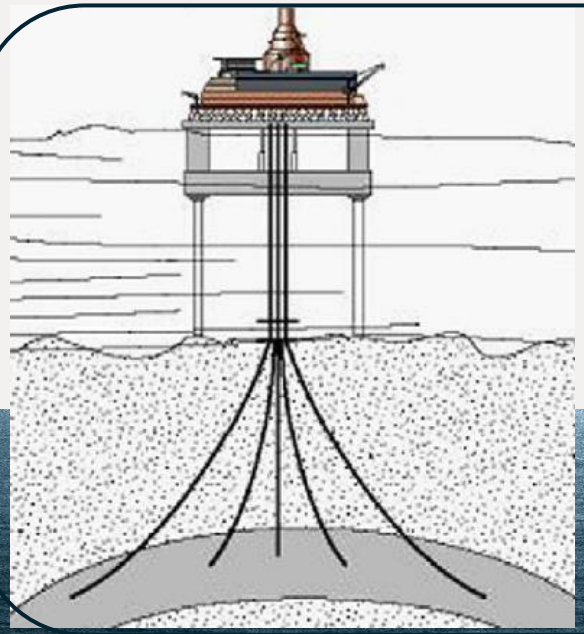
- Oil and gas operations in water depth less than 100 meters (~328 feet)

Deepwater

- Defined by NORSOK D-010 as water depth greater than 600 meters
- Deep water may range from 600 m to 3000 m and ultra-deep water beyond this depth.

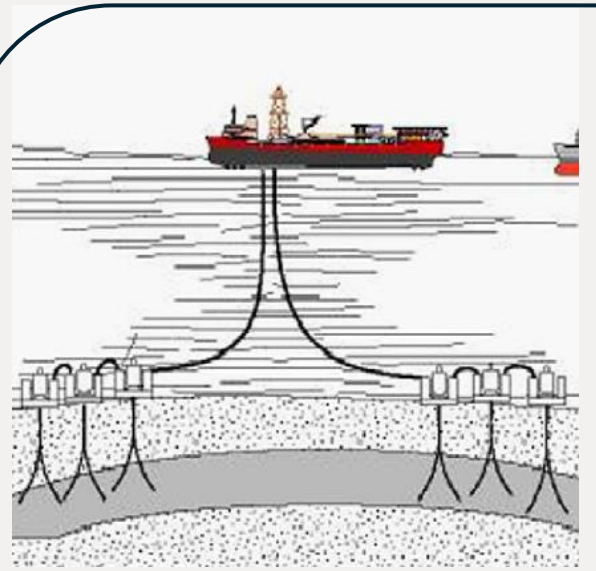
OFFSHORE VARIANT TO DEEPWATER





DRY TREE

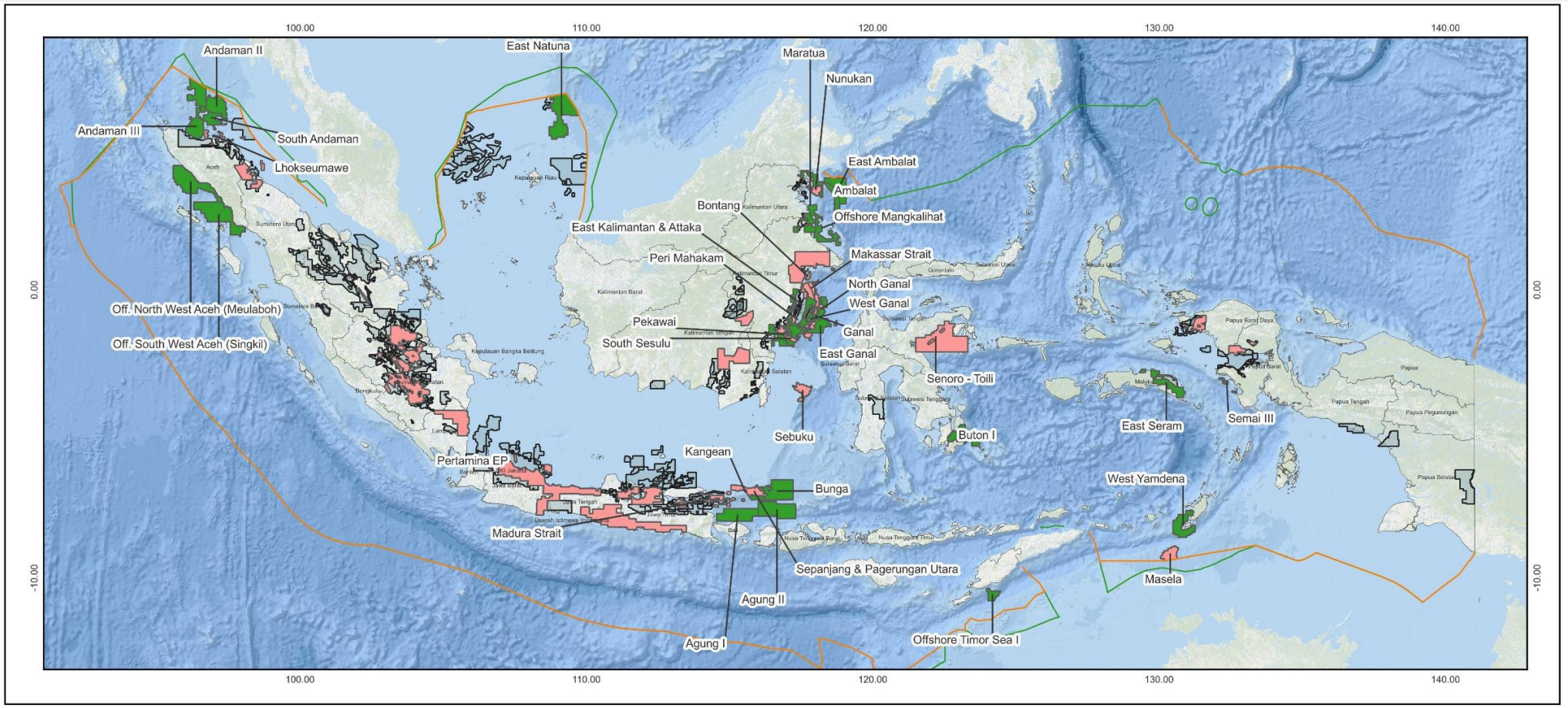
- Direct access for human
- Onshore location
- Hydraulic or manual control equipment



WET TREE

- Remote control
- Offshore location
- Electro-hydraulic control equipment

Indonesia Offshore Map



N

0 500 1.000 km

Landas Kontinen

ZEE

WK Migas Deepwater (>200m)

WK Eksploitasi

WK Eksplorasi

WK Migas Indonesia Lainnya (Onshore / <200m)

Batas Administrasi Indonesia

ESRI Ocean

Status 1 September 2023

Deepwater Development in Indonesia

(Costamte, et al., 2021; Proceedings, IPA 45th Convex, 2021)

Field/Project	Basin	Discovery	Final POD	On-stream	Discovery to Final POD	Discovery to on-stream	X WD (m)	X MD (m)
West Seno Field	Kutai	1998	1999	2003	1	5	1.024	3.764
Bangka Field	Kutai	1999	2008	2016	9	17	970	3.173
Jangkrik Field	Kutai	2009	2011	2017	2	8	387	2.544
Jangkrik NE Field	Kutai	2011	2013	2017	2	6	418	3.480
Merakes Field	Kutai	2014	2018	2021	4	7	1.453	2.465
Jambu Aye Utara Field	North Sumatera	1986	2012	not yet	26	-	663	2.950
IDD Project								
- Bangka Field (Produced)								
- Gada Field		2000					1.898	3.847
- Gendalo Field	Kutai	2000	2024	2028	-	28	1.348	3.992
- Maha Field		2002					742	4.097
- Gandang Field		1999					1.709	3.570
Abadi Field	Timor	2000	2023	2030	19	30	579	4.016

Offshore vs Onshore: Complexity Drivers



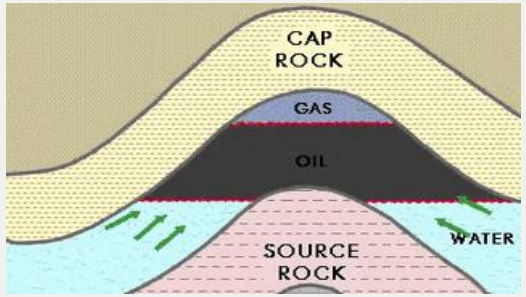
Environment

- Waves, currents, wind
- Weather uncertainty
- Metocean variability



Subsurface & Seabed

- Shallow gas & hydrates
- Soil instability
- Geohazards



Equipment & Technology

- Subsea BOP & riser system
- Dynamic positioning
- ROV dependency



Operations & Logistics

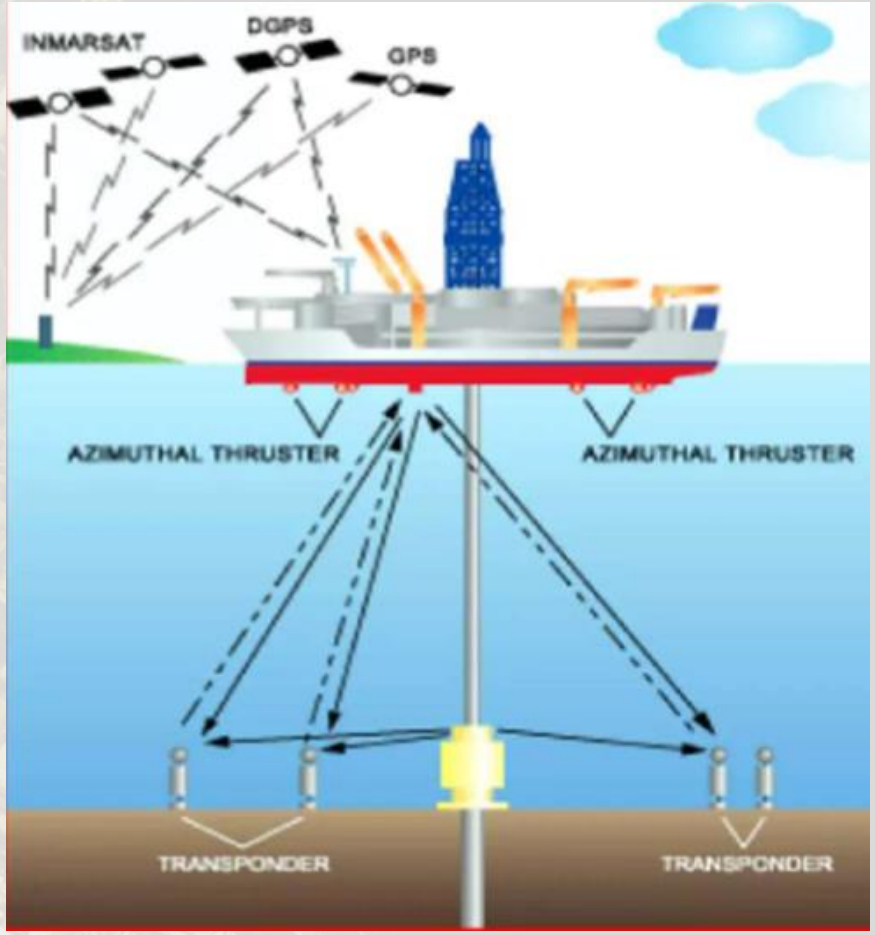
- Limited access
- Vessel & rig availability
- High cost of delay



Deepwater Technology #01: Rig Dynamic Positioning System

- GPS satellites and acoustic positioning system communicate with transponders on the ocean floor

- Propulsion systems counteract the forces acting on the ship e.g.
  - Wind, wave, current direction and speed.



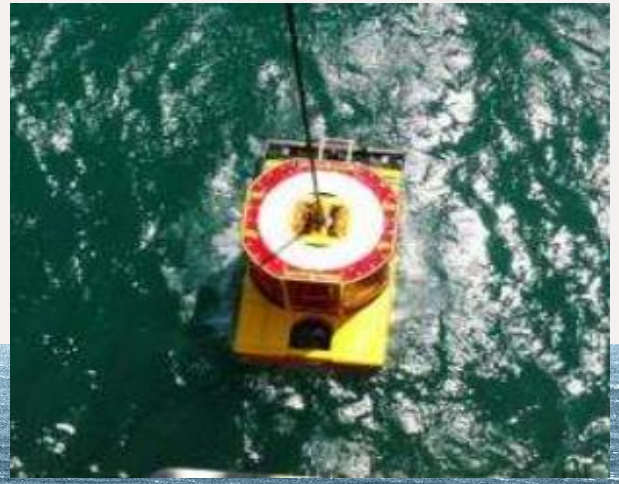
Deepwater Technology #02:
ROV (Remote Operated Vehicle)





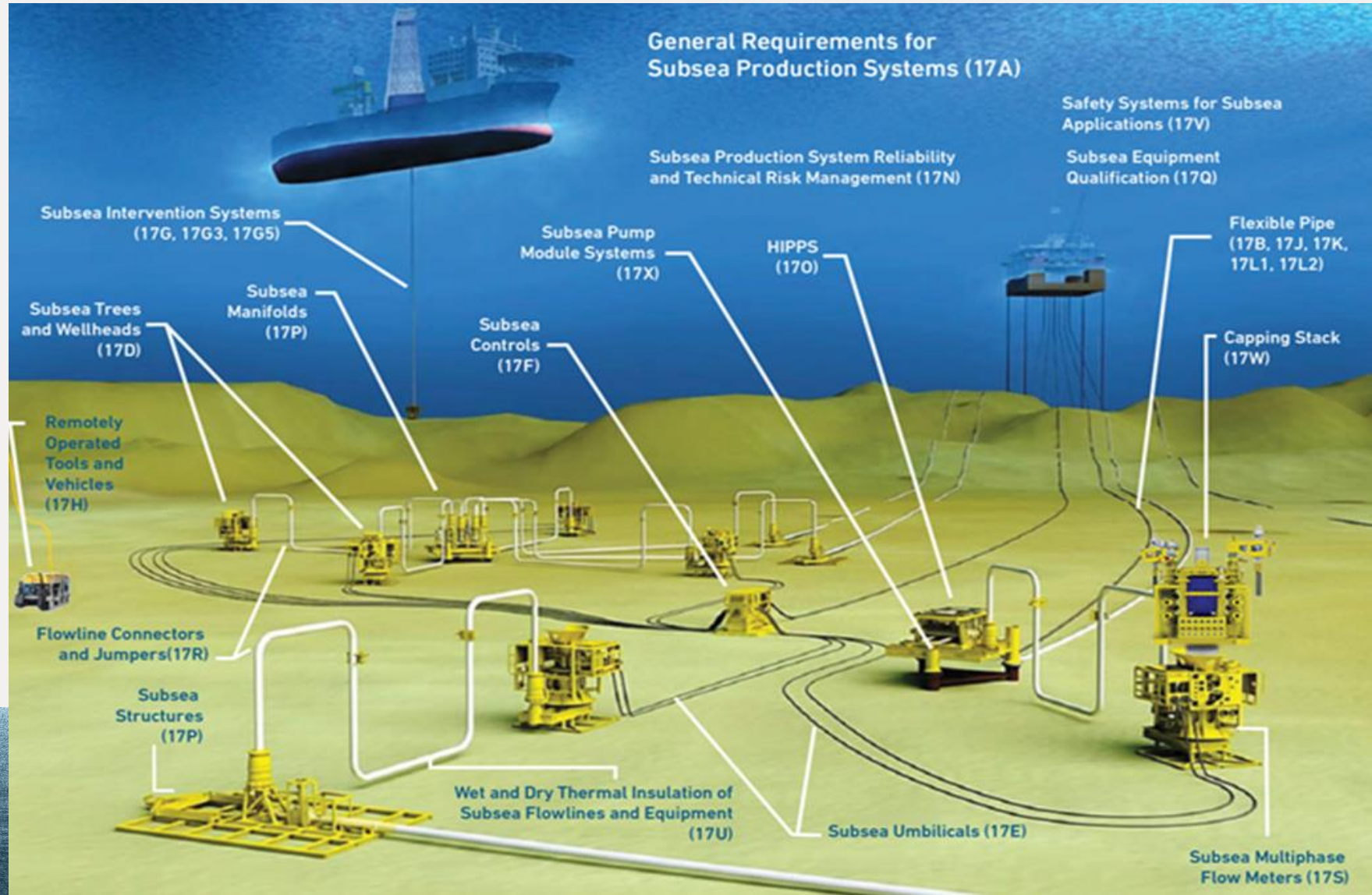






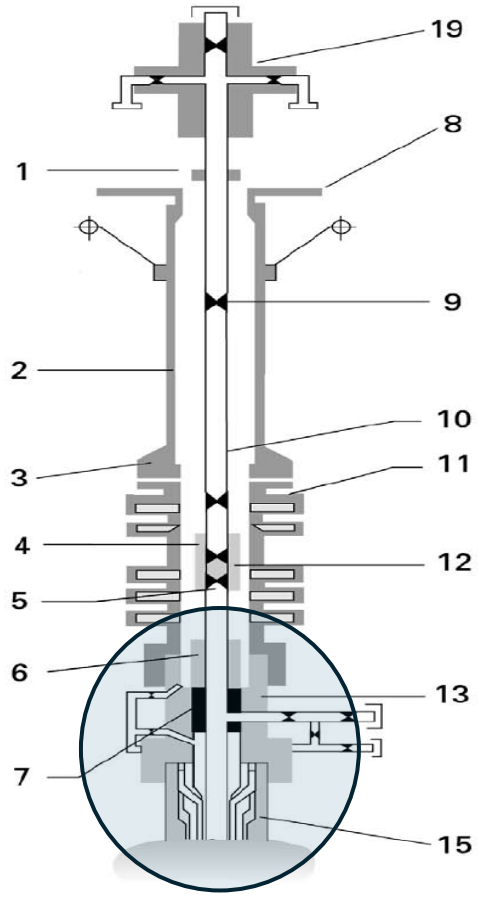
Deepwater Technology #03: Subsea Equipment

1. Subsea Hardware: Wellhead, X-Tree, Manifold, Structure, Connector, and Pump
2. Flowlines and Risers: Flexible pipe, flowline connector and jumper
3. Control Systems: Umbilical, production control system, SUT (Subsea Umbilical Termination)
4. Intervention Systems: Intervention Workover Control System (IWCOS), TBIRS Through BOP Intervention Riser System (TBIRS), ROV, and Subsea Capping Stack

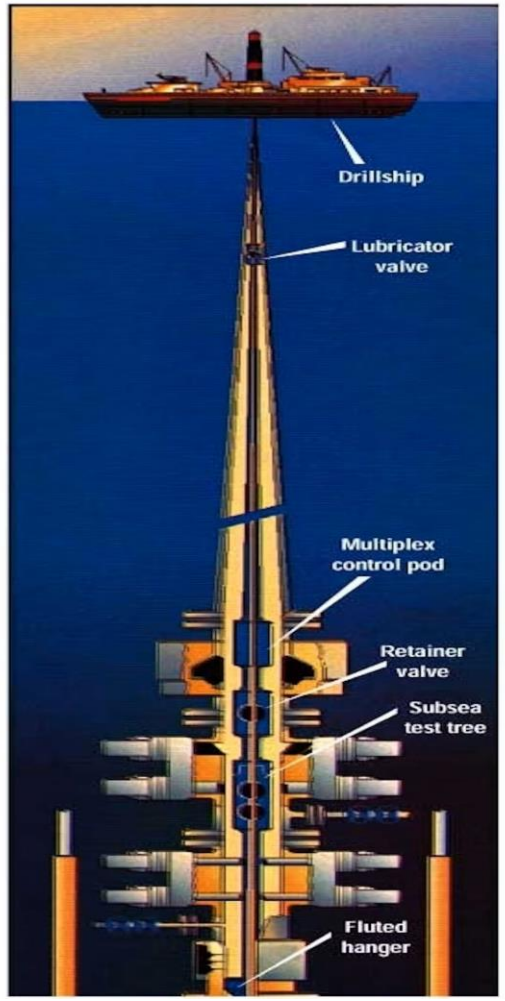


Deepwater Technology #04: Detailed Subsea Intervention System

Subsea Tree and Riser System



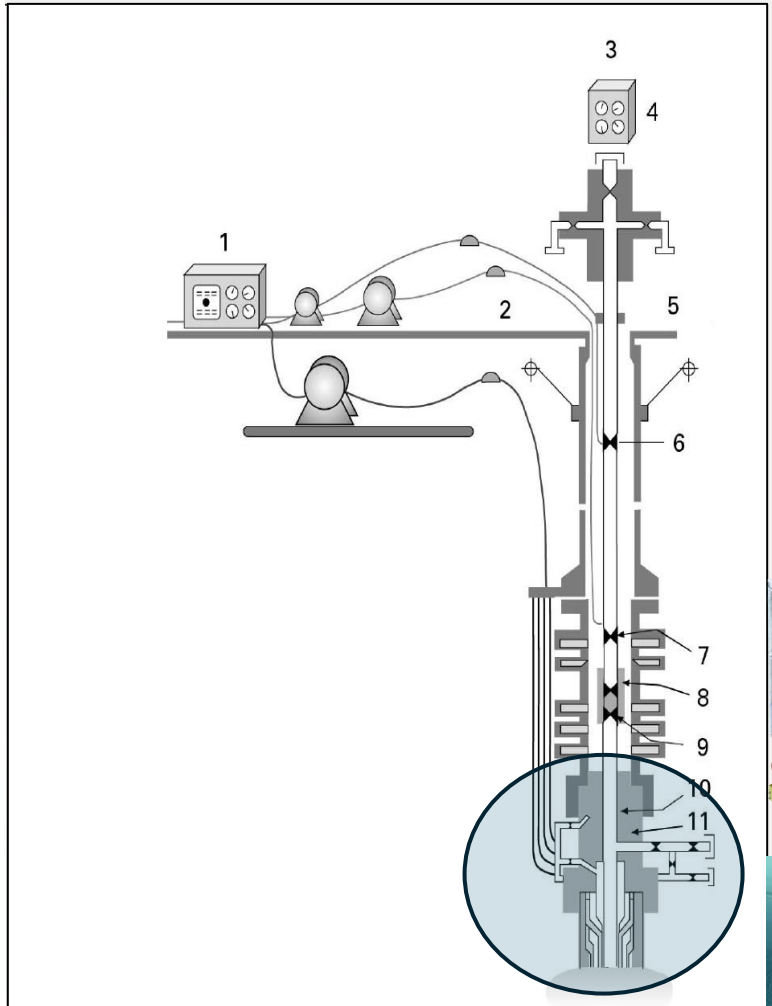
a) Horizontal tree



Key

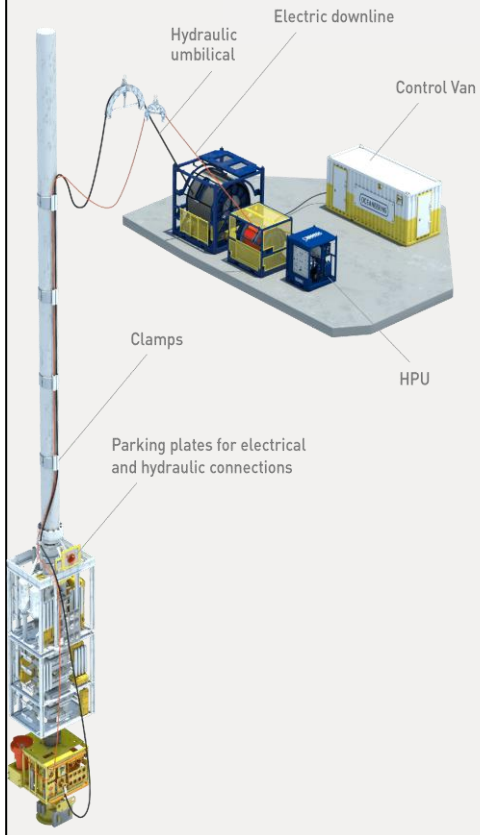
1	swivel (optional)	8	drill floor	14	TH
2	marine riser	9	lubricator valve	15	wellhead
3	flex joint	10	landing string	16	workover riser
4	EDP	11	BOP annular bag	17	riser stress joint
5	cutter valve	12	subsea safety tree rams	18	EDP/LRP
6	TH running tool	13	tree	19	surface tree
7	TH				

Installation Workover Control System



Key

1	HPU and control panel	7	retainer valve
2	swivel (optional)	8	EDP
3	local control panel	9	cutter valve
4	surface tree	10	TH running tool
5	drill floor	11	TH
6	lubricator valve		





Deepwater Drilling and Completion (D&C)
Example of Typical Well Design

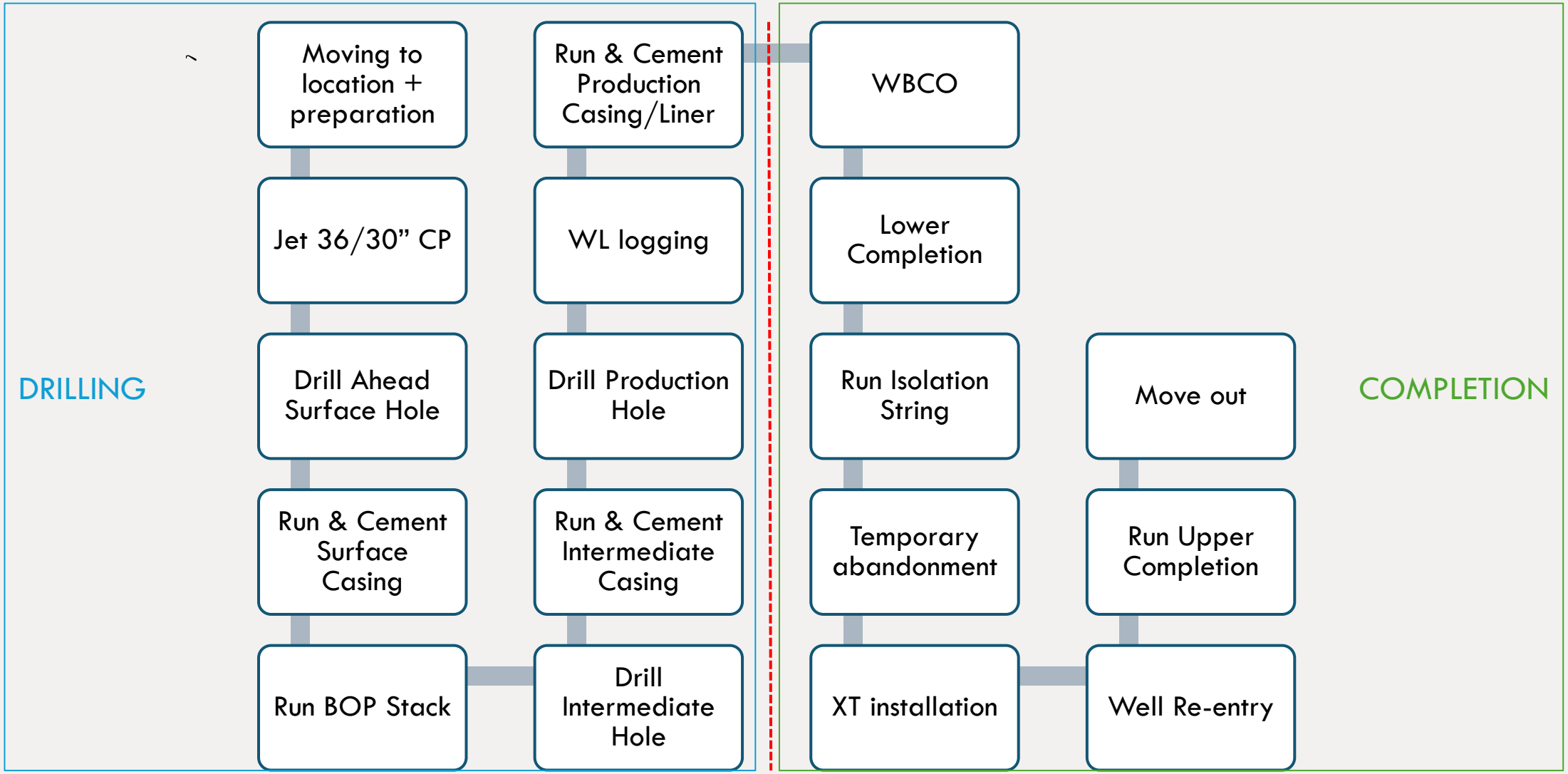
Typical Drilling Well Schematic

Typical Completion Well Schematic

Well design:

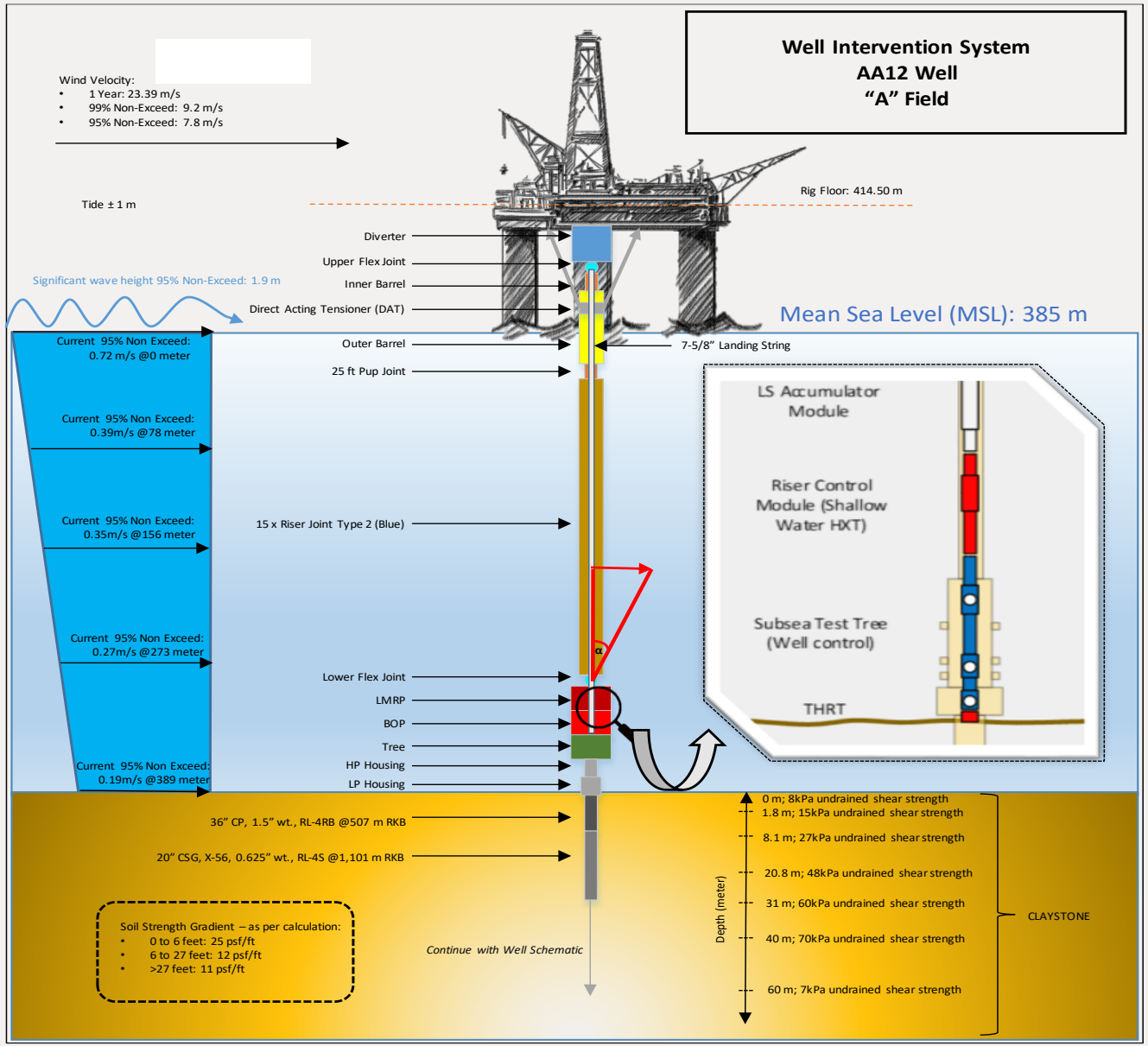
Intentionally removed for sharing material

Deepwater Drilling and Completion (D&C) Example of Typical Sequences & Methods



DEEPWATER CASE #01: SUBSEA ENVIRONMENT

Reference: SPE-215177-MS
"A Success Story of Using Deepwater Semisubmersible Drilling Rig for Workover Operation in 400 Meters Water Depth without Utilization of Anchoring System"



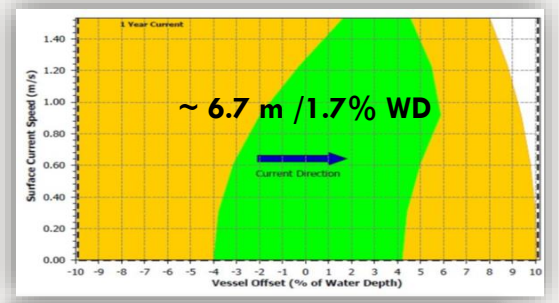
INPUT
- System Set-up : Onboard and TBIRS Specification
- Environment Condition (wave, current, wind and soil condition)
- Conductor bearing capacity confirmations

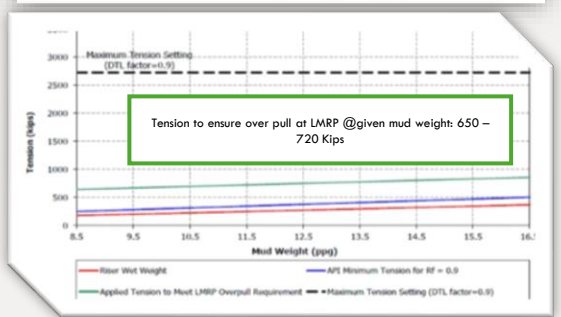
ANALYSIS
- TBIRS/C-WO Riser Analysis (API-17G)
- Drilling Riser Analysis (API 16Q)
- Time Series Load Analysis of TBIRS

OUTPUT
- Static Operating Condition (circle & drift off management)
- Vessel Heading

EXECUTION
- Rig Power Management
- Heading Management
- ERA-ARA Monitoring
- C-node transponder
- TBIRS – ESD Timing
- UDS in LMRP

Drilling Riser Analysis – API 16Q





TBIRS (C/WO Riser) – API 17G

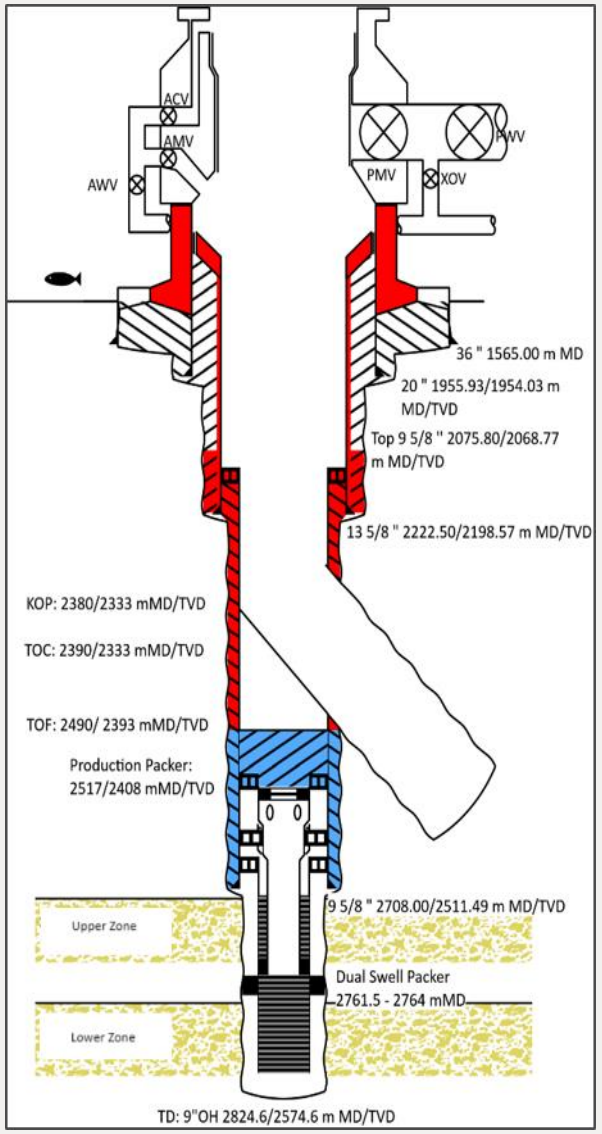
No.	Case during operations in shallow water	Maximum Angle	Rig Drift	Mass Structure to consider	Limit by
		Deg	m		
1	Tubing hanger Retreival & Installations (TBIRS - Running and retreival)				
a	Component Running Through UFJ	2.41	16.8	TBIRS + 7 5/8" 60 ppf String #P110 + CTLF + Tubing String Weight (as relevant)	*Limit by TBIRS's LSAM and connection in Tubing hanger running tools & its adaptor*
b	Running Through Lower Flex Joint	2.48	16,7	TBIRS + 7 5/8" 60 ppf String #P110 + CTLF + Tubing String Weight (as relevant)	Limit by connection in Tubing hanger running tools & its adaptor
	Note: 1. Running speed = 0.5 m/s, Pressure in riser system = Hydrostic. 2. String Weight is assume to be 74 MT				

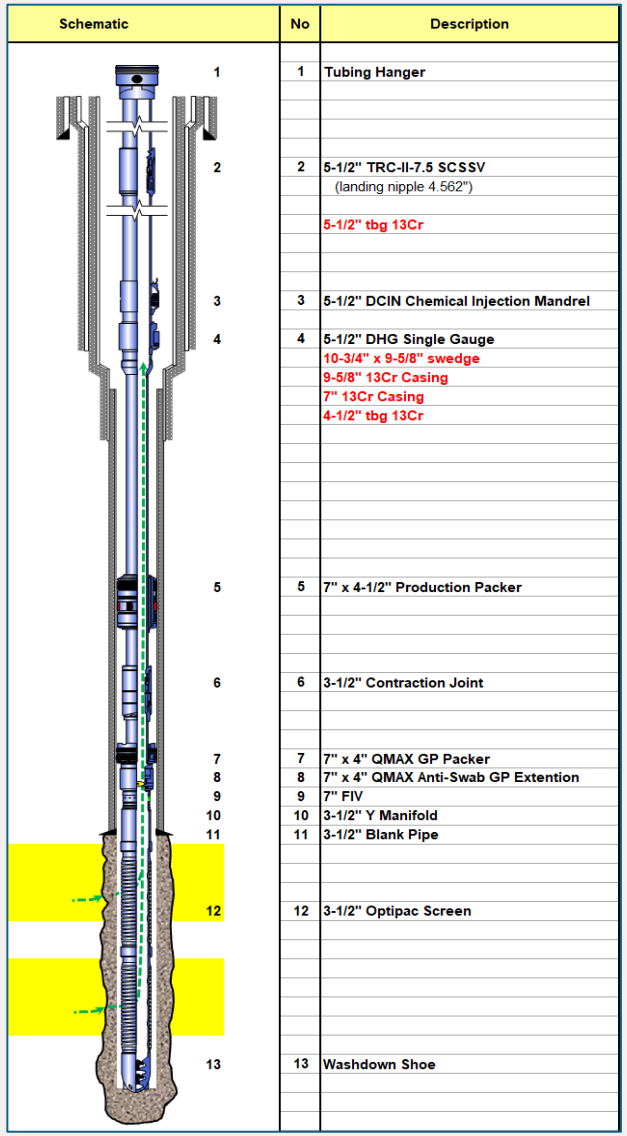
Time Series Load Analysis

All Sensitifity Case										
Component	Locations Relative to MSL (m)	Sofft Soil	Stiff soil	Very stiff soil (Fix at mud line)	Decrease Over-pull (-25%)	Increase Over-pull (+25%)	Soft solid Head Sea	Soft Soil Quarter sea	Modified Riser Tally	Wave Spreading
	m	years	years	years	years	years	years	years	years	years
X-tree Connector	-385.48	1.19	1.39	6.9	5.69	1.36	3.06	51.68	1.17	3.01
Wellhead Hub	-385.48	2.18	2.42	8.84	7.62	2.34	5.07	54.98	2.12	2.12
BOP Hub/Connector	-382.965	10.22	4.73	5.32	13.67	10.85	28.74	378	9.8	9.8
Otis Connections	40.37	0.67	0.67	0.67	0.69	0.69	2.29	2.57	0.69	1.49
Tubular Riser Connection -1	-372.48	2.41	20.96	3.18	1.03	6.91	4.99	340	2.22	6.56
Tubular Riser Connection -2	-39.55	4.73	4.52	4.58	4.47	4.69	4.88	5.32	6.68	8.83

DEEPWATER CASE #02 : SIDETRACK & RECOMPLETION

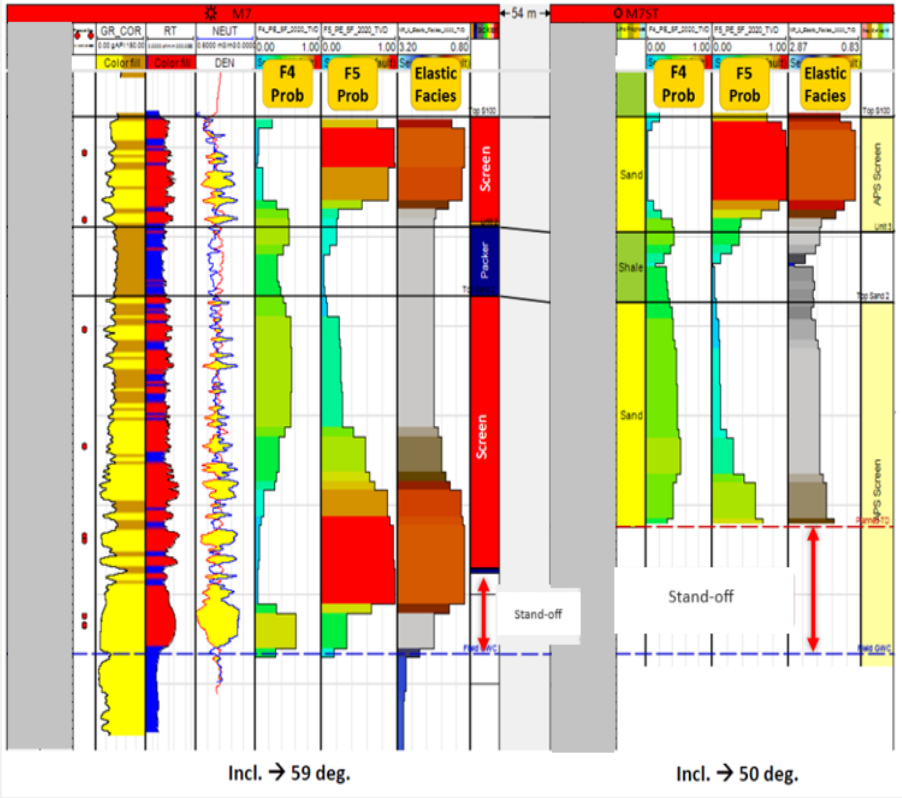
Reference: SPE-217910-MS
"Maximizing Slim Hole Well Deliverability: First Installation of 3.5" Alternate Path Open Hole Screen in Indonesia and Implementation of Erosional Velocity Analysis"





No	Description
1	Tubing Hanger
2	5-1/2" TRC-II-7.5 SCSSV (landing nipple 4.562")
	5-1/2" tbg 13Cr
3	5-1/2" DCIN Chemical Injection Mandrel
4	5-1/2" DHG Single Gauge
	10-3/4" x 9-5/8" swedge
	9-5/8" 13Cr Casing
	7" 13Cr Casing
	4-1/2" tbg 13Cr
5	7" x 4-1/2" Production Packer
6	3-1/2" Contraction Joint
7	7" x 4" QMAX GP Packer
8	7" x 4" QMAX Anti-Swab GP Extention
9	7" FIV
10	3-1/2" Y Manifold
11	3-1/2" Blank Pipe
12	3-1/2" Optipac Screen
13	Washdown Shoe

- Perform workover and sidetrack from 9-5/8" casing on M6 and M7 wells
- Complete M6ST and M7ST wells with 6"x8" OHGP with regular upper completion considering stand off from GWC and comingle flow from both zones.
- Designed for high flow despite of small bore.



Screen interval

Upper Zone (UZ)

Screen interval

Lower Zone (LZ)

DEEPWATER CASE #03 : Well Stimulation (Acidizing)

Reference: IPTC-24112-MS
"Acid Stimulation Challenge on Subsea Gas Wells with Depleted Reservoir Pressure"



Vessel Availability
- ***No LWIV*** *available in the region*
- ***Semisubmersible rig*** *was* ***nearby*** *& used in previous sidetrack campaign in "A" field.*

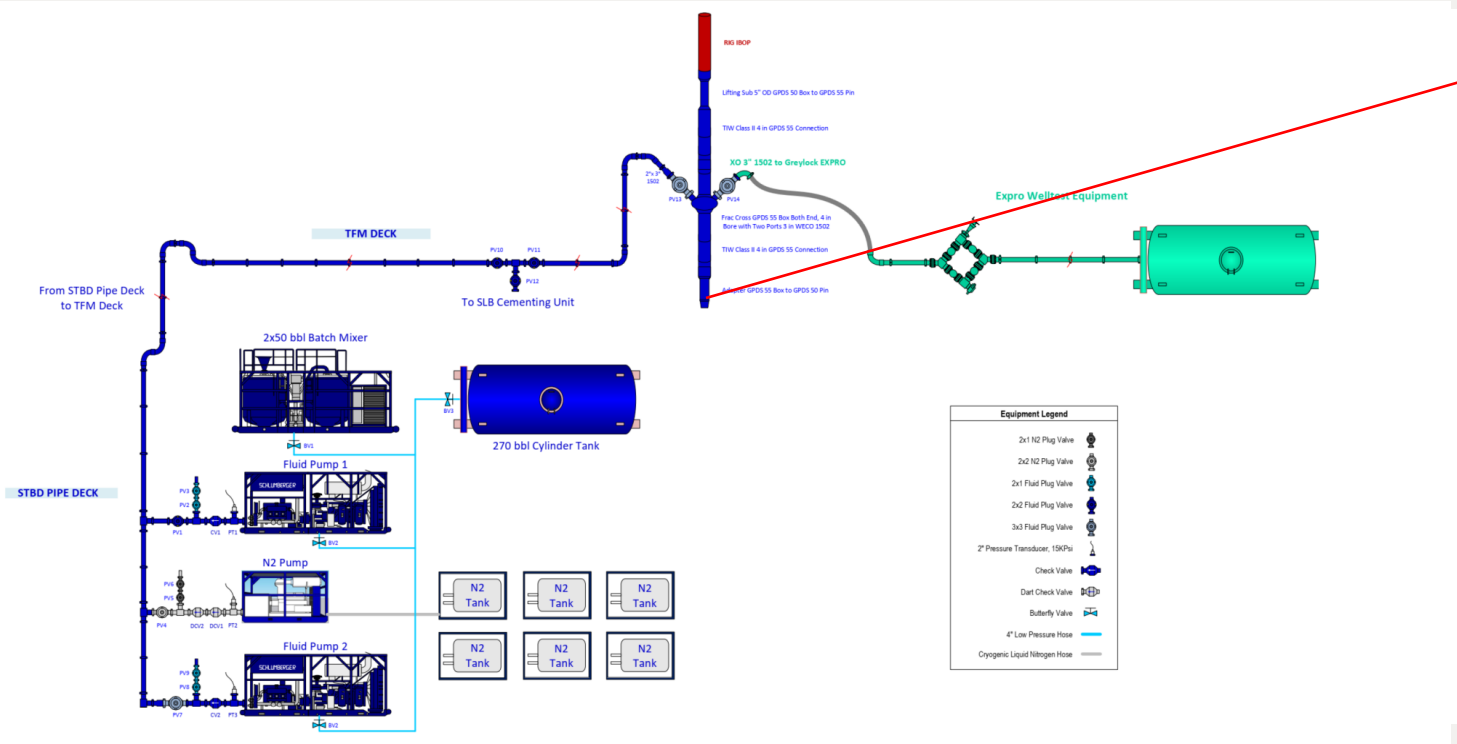


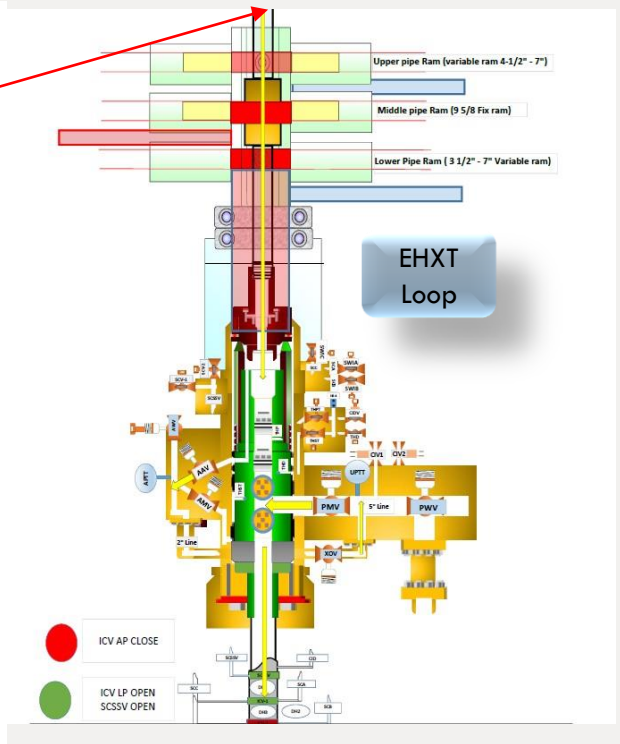
Intervention Technique
- ***Bullhead*** *via annulus of horizontal* ***X-Tree Loop***
- ***No Crown Plug retrieval*** *using Coiled Tubing*



Monitor and Control
- *Disconnect wells from field cluster*
- *Deploy* ***IWOCS*** *to monitor and control.*

	Typical One Zone Acidizing Operation
1	Pump nitrified fluid and fill up the tubing
2	Continue fill up the tubing
3	Formation started to receive liquid. 5% NH4Cl brine is entering the formation
4	HCl 15% pre-flush started entering formation and react with the scales, shown by decreased in DHG 1 reading
5	13.5:1.5 HCl-HF acid started entering formation
6	HCl 15% post-flush started entering formation
7	5% NH4Cl started entering the formation
8	No liquid is left on the tubing





Key Technical Competence in Deepwater Project

Basic knowledge of offshore well engineering and operation including drilling, completion, workover, well intervention, subsea, and engineering application.

Understanding of Standard: API, NORSOK, Company Specific Guideline

Certified in BOSIET (offshore safety) and Well Control (IWCF, IADC, etc.)

Familiarity with administrative work: coordination with authorities, programs, budgeting, reporting, etc.

Developing self-awareness and staying updated with new technologies through conference events, papers in website, and service providers presentation.

Footage of Deepwater Rig



- Video taken from Leo Kencono Youtube Channel
- Leo Kencono – HSE Spv Eni at SSV Catarina on 2022

Semisubmersible Rig - Deepwater Drilling & Workover

Thank You!

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